
Learning to See Sharper

Deep Learning for Astronomical Data Enhancement

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The space astronomical data live in

FRAMING

Coverage — *how much you see*

the whole sky · the full EM spectrum · a long time baseline

Resolution — *how finely you see it*

pixel scale / PSF · $R = \lambda/\Delta\lambda$ · observing cadence

Depth — *how faint you reach*

higher S/N on the bright — and reach to the faintest, furthest objects

EVERY FACILITY BUYS ONE AXIS WITH ANOTHER

Rubin / LSST

the whole southern sky every few nights — at $\sim 1''$ seeing

Euclid

14,000 deg² at $0.1''$ in the visible; $0.3''$ in the near-IR

JWST

$0.06''$ and exquisite depth — over a vanishingly small field

WISE

all-sky at $3.4\text{ }\mu\text{m}$ — with a $6''$ beam

Slitless / prism spectra

millions of galaxies at once, at $R \sim 100$

No single dataset fills this space. **But surveys overlap — and where they overlap, one survey has already recorded what the other is missing.**

The inverse problem: why sharper is hard

- What we record is a degraded version of the sky: convolved with the PSF, sampled onto pixels, buried in noise.
- Inverting that is **ill-posed** — the solution is neither unique nor stable. Very different skies can produce the same data.
- Classical deconvolution — Richardson–Lucy, Wiener, Tikhonov, wavelet methods — has carried the field a long way, but it **plateaus** once the resolution element approaches the separation between the features you care about.
- What is missing is a **prior**: a model of what real galaxies and real spectra actually look like. Hand-tuned priors are the ceiling of the classical approach.

$$\mathbf{d} = \mathbf{A} \mathbf{x} + \mathbf{n}$$

$\mathbf{d}$ what we observe

$\mathbf{A}$ PSF blur · pixel sampling · instrument

$\mathbf{x}$ the sky we want

$\mathbf{n}$ noise

THE RESTATEMENT

Don't invert the degradation by hand. Learn the mapping directly, from real examples of the same objects seen both ways.

Survey overlaps are the prior – and they are free

- Wherever two surveys see the same sky, every object is a **training pair**: the same galaxy, recorded coarsely and recorded sharply.
- No simulation, no hand-built forward model, no labels. The pairs **are** the prior — real galaxies teaching the network what plausible structure looks like.
- Train on the overlap; apply to the whole archive. The overlap is small. The archive is not.

WHAT THIS TALK IS ABOUT

Five measurements, five architectures. Matching the model to what the measurement actually needs — not the other way around.

THE PAIRS IN THIS TALK

Subaru HSC → HST

0.17" from the ground → 0.08" from space

WISE → Spitzer

6.1" all-sky → 1.7" pointed

Euclid NISP → JWST NIRCams

0.3" wide-field → 0.06" deep

HST & JWST → Euclid, Roman

sharp truth → survey realism

NIRSpec prism → gratings

R ~ 100 → R ~ 1000

Five problems, five right-sized architectures

ROADMAP

1 • Deblending

Subaru HSC → HST

A super-resolution GAN separates galaxies that ground-based imaging merges into one. Blending is a leading weak-lensing systematic.

*GAN
Hemmati et al. 2022*

2 • The infrared sky

WISE → Spitzer

4.6× super-resolution of the all-sky 3.4 μm archive toward the resolution Spitzer reached only where it pointed.

*Enhanced RCAN
Rezaee et al. 2026*

3 • Shapes in the near-IR

Euclid NISP → JWST

5× super-resolution of NISP Y, measured against NIRCам: how much shape information is genuinely recoverable?

*ResNet / RRDB
Everetts et al. 2026*

4 • Simulation

*HST & JWST → Euclid,
Roman*

Diffusion models generate galaxy populations with real morphologies for shear calibration, where analytic profiles fall short.

*DDPM
Scognamiglio et al. 2025, 2026*

5 • Spectral super-resolution

prism → grating

A physics-informed network resolves emission lines that are blended beyond recovery at prism resolution.

*Attention + line model
Haghjoo et al. 2026*

Two strands — **imaging** (1–4) and **spectroscopy** (5) — and one argument running through both: understand the measurement first, then choose the network.

PART ONE

Imaging

Separating what is blended, sharpening what is coarse, measuring shapes where we could not before — and generating galaxy populations realistic enough to calibrate against.

Blending: a systematic that grows with depth

1 • DEBLENDING

- At the depths Rubin will reach, up to a quarter of sources in first-year ground-based stacks — and up to half at full depth — are blended at seeing-limited resolution (*Chary et al. 2020*).
- Two galaxies read as one: the centroid is wrong, the flux is wrong, and the shape is wrong. For weak lensing, the last of those is the expensive one.
- A controlled test: pairs of real CANDELS galaxies (HST F775W, $0.06''/\text{px}$, $0.08''$ FWHM) placed at random separations and orientations, then degraded to Subaru HSC sampling — so the truth is known for every blend.
- **Even at HST resolution, ~20% of those blends are unrecognizable. At HSC-like resolution, ~90%.**

THE QUESTION

How much of the resolution the atmosphere took away can a learned prior put back — and does it put back the right thing?

~20%

unrecognizable even at
HST resolution

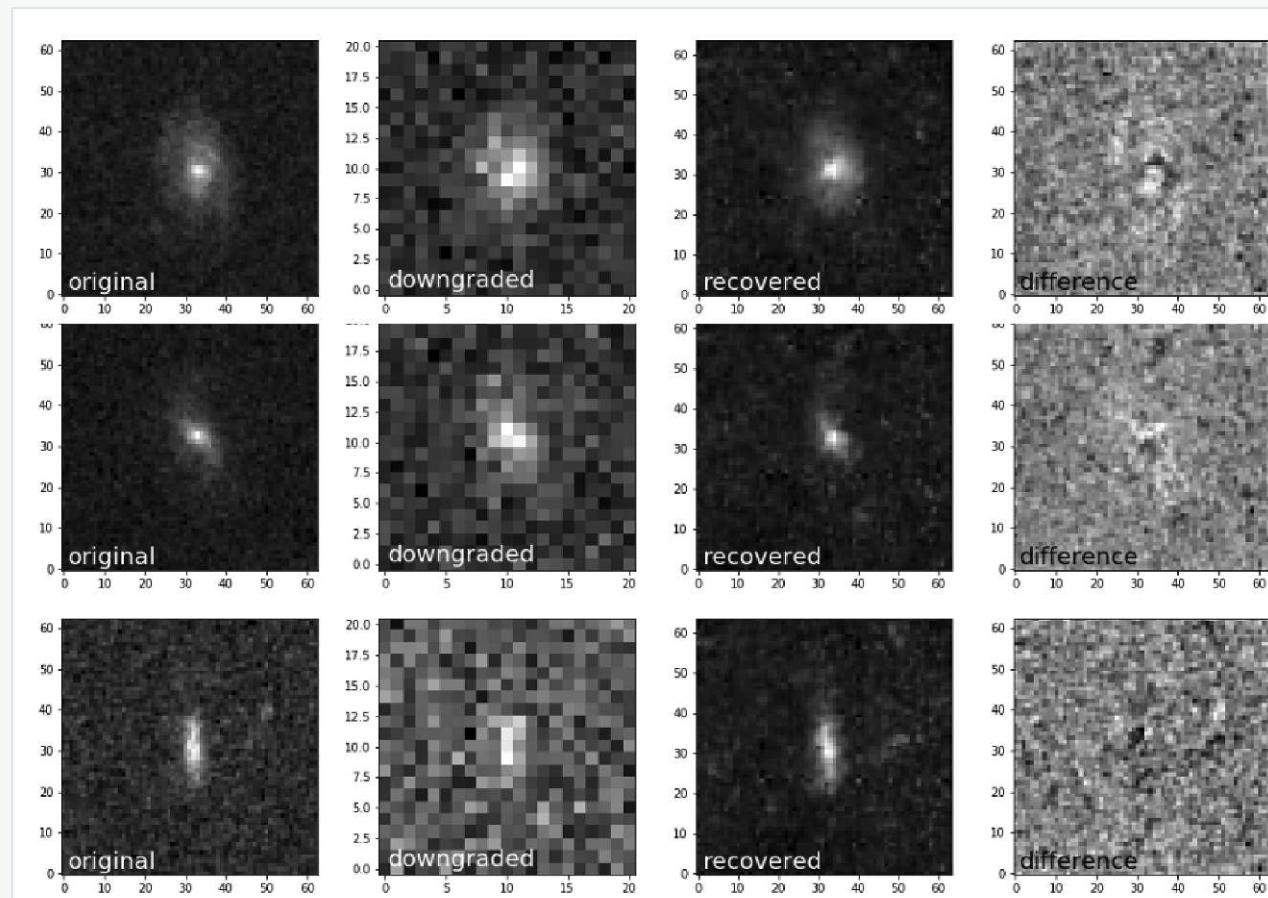
~90%

unrecognizable at
HSC-like resolution

Blending is not a nuisance at the edge of the error budget. For Stage IV weak lensing it is **one of the leading systematics** — and it is a resolution problem.

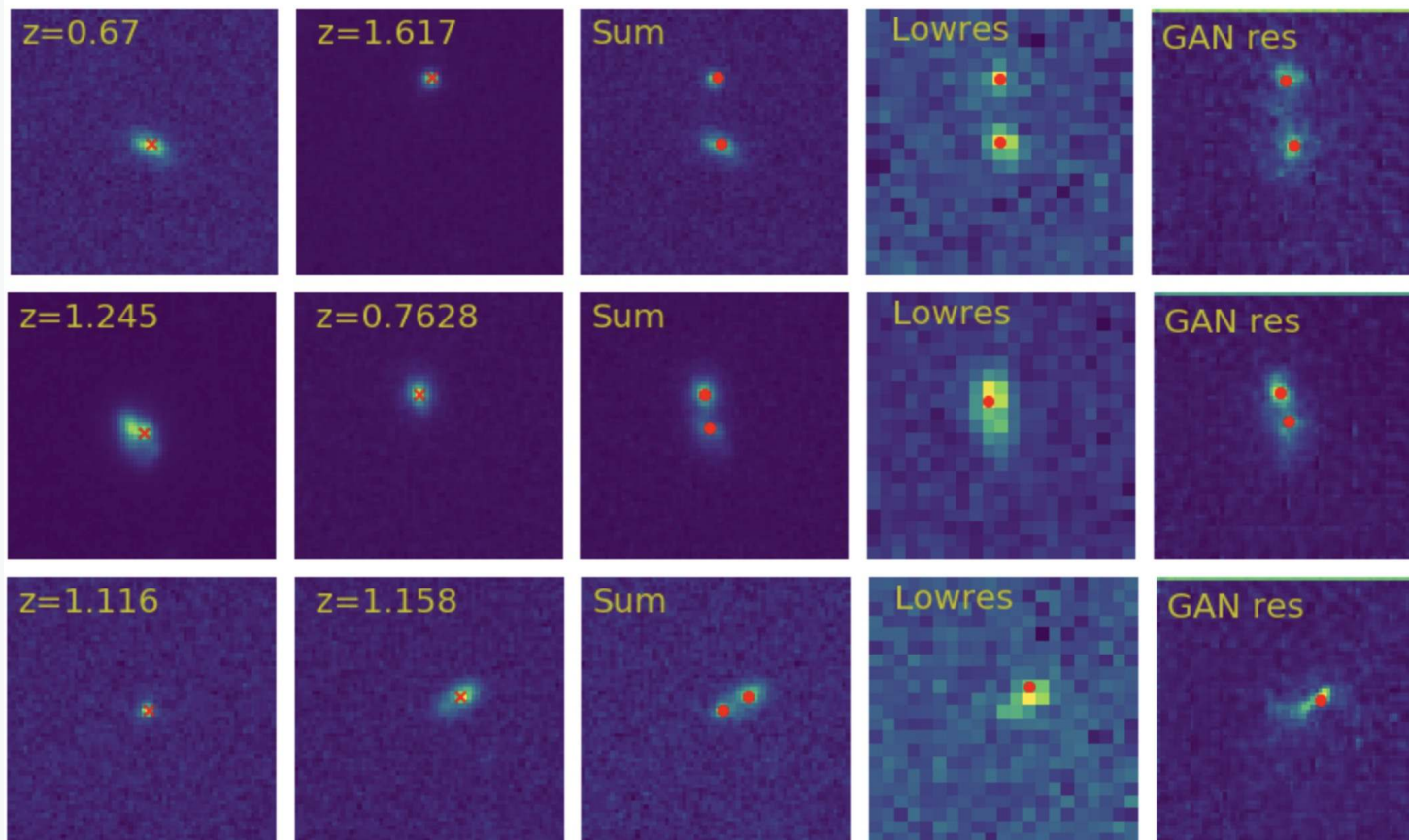
Building the deblender: two GANs

- Start unconditioned: a vanilla GAN trained on CANDELS turns a noise vector into an HST-resolution galaxy. Proof that the network has learned the manifold of real morphologies.
- Then condition it — feed the degraded HSC-like cutout instead of noise. Same adversarial training; now the generator must produce the HST version of *this* galaxy.
- 4–5 convolutional layers in generator and discriminator, PyTorch, ~4 hours on three GPUs, a few thousand galaxy images.
- **A small network on ordinary hardware. The leverage is in the pairs, not the parameter count.**



Hemmati et al. 2022 — original HST · degraded to HSC sampling · GAN recovery · difference

Finding both peaks



Columns 1–2: the two input galaxies. 3: their sum at HST resolution. 4: degraded to HSC sampling — the GAN input. 5: GAN output. Red dots mark peaks found by the same detector in each.

- Each blend is two real CANDELS galaxies, moved to a random off-centre position and orientation, summed, then degraded.
- Run the same peak finder on all three versions. At HSC-like resolution it finds one source where there are two; on the GAN output it finds both.
- The GAN is not asked to preserve flux — the recovered **positions** are what gets used, measured back on the original low-resolution image.

90% → 50%

unrecognizable blends: HSC-like input versus GAN output

~1%

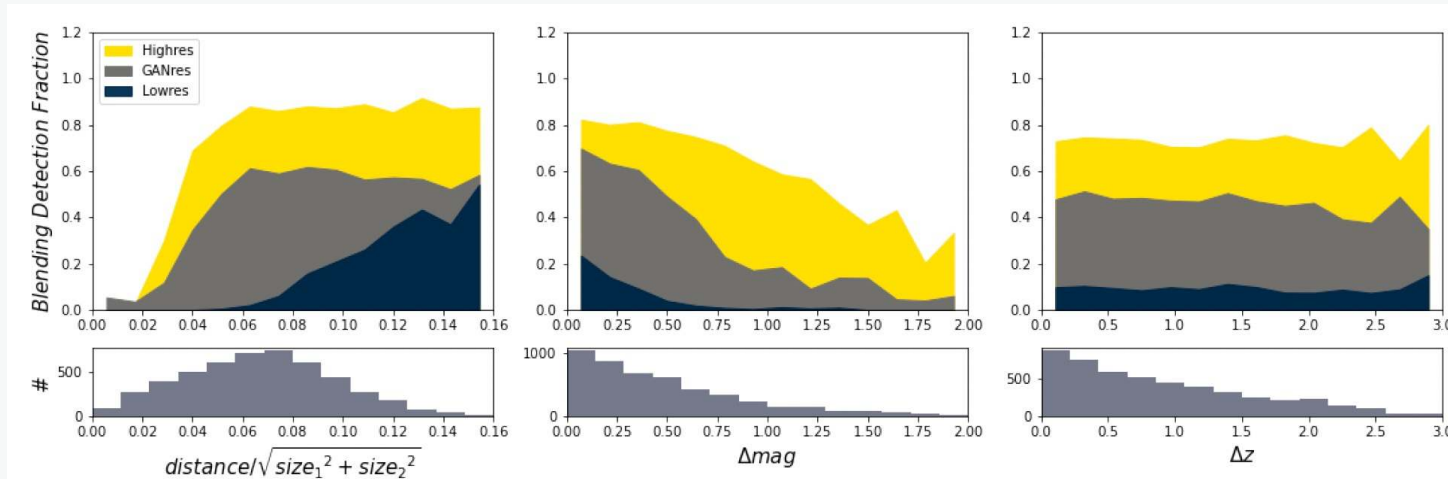
false-positive rate — the GAN is not inventing companions

both peaks

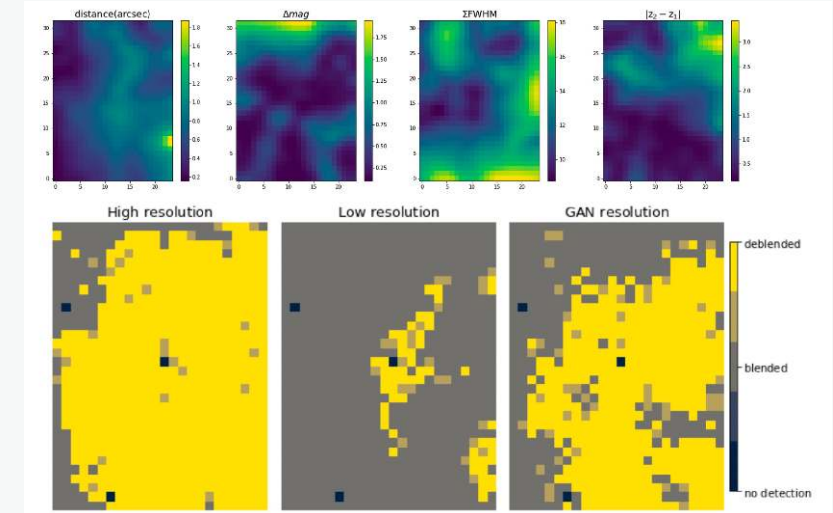
recovered where the low-resolution image showed only one

Where it works – and what it buys

1 • DEBLENDING



Fraction of blends whose two peaks are recovered, versus separation (scaled by size), magnitude difference, and redshift difference. Yellow: HST truth. Grey: GAN. Dark blue: HSC-like input.



Hemmati et al. 2022 — map of blend parameters (top); per cell, whether both peaks are found at HST, HSC-like and GAN resolution (bottom).

~10×

better photometry for blended objects,
using the positions the GAN recovers

≈10% more

blends detected by the 7-band
multi-wavelength GAN

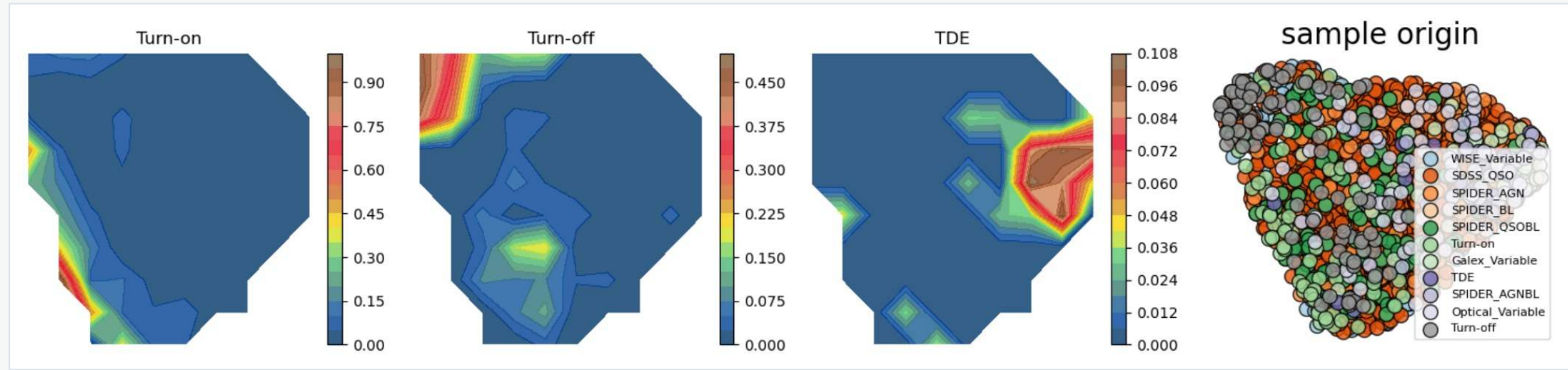
7 bands

optical + NIR: the same architecture,
one extra input dimension

Why it matters: recovered positions feed straight into photometry and shape measurement — the two places blending does its damage.

WISE sees the physics – through a 6" beam

2 · WISE → SPITZER



Hemmati et al. 2026, ApJ 998, 130 — a UMAP manifold of WISE W1 light curves for ~2,000 AGNs at $z < 1$, learned without labels. Each panel: where one labelled class lands on the manifold (colour = local fraction of that class). Right: every source by catalogue of origin.

- NEOWISE revisits every point on the sky twice a year: a **decade-long 3.4 μm light curve for every AGN in the sky**, already sitting in the archive.
- A manifold learned from those W1 light curves alone — **no labels in training** — separates turn-on from turn-off changing-look AGNs and isolates tidal disruption events (Hemmati et al. 2026; ~2,000 AGNs at $z < 1$). In 65,000 type-2 AGNs the same coordinates track [O iii] luminosity, stellar mass and D_n4000 .
- Every one of those light curves is measured through a **6.1" beam**: 12 kpc at $z = 0.1$, 38 kpc at $z = 0.5$ — the whole host galaxy, and often a neighbour.

WHY THIS IS A RESOLUTION PROBLEM

Selecting this autumn's spectroscopic follow-up targets from that manifold: one candidate in three has a Pan-STARRS neighbour inside the WISE beam, and 75 of 1,061 have one bright enough to contaminate the W1 light curve. The variability is real. The beam cannot say where it comes from.

The all-sky infrared archive is blurry

2 · WISE → SPITZER

- WISE mapped the **entire sky** at $3.4\text{ }\mu\text{m}$. W1 has a PSF FWHM of $\approx 6.1''$ at $2.75''/\text{px}$, reaching $\approx 54\text{ }\mu\text{Jy}$ (5σ) in AllWISE — with a decade of time-domain coverage through NEOWISE.
- Spitzer/IRAC Ch1 reached $\approx 1.7''$ FWHM at $0.6''/\text{px}$ at almost the same wavelength ($3.6\text{ }\mu\text{m}$) — but only where it pointed, and it stopped observing in 2020.
- COSMOS has both. $\sim 390,000$ paired cutouts drawn uniformly across the overlap, with a held-out test set of 83,592 on which every number is reported.
- An Enhanced RCAN — residual channel attention — trained with a source-focused loss: the model is graded on the pixels that belong to real sources, not on reproducing blank sky.

THE OPPORTUNITY

Spitzer-class resolution is a property of a few selected fields. WISE coverage is a property of the whole sky. A learned mapping between them travels.

4.6x

spatial super-resolution

W1 → IRAC Ch1

390k

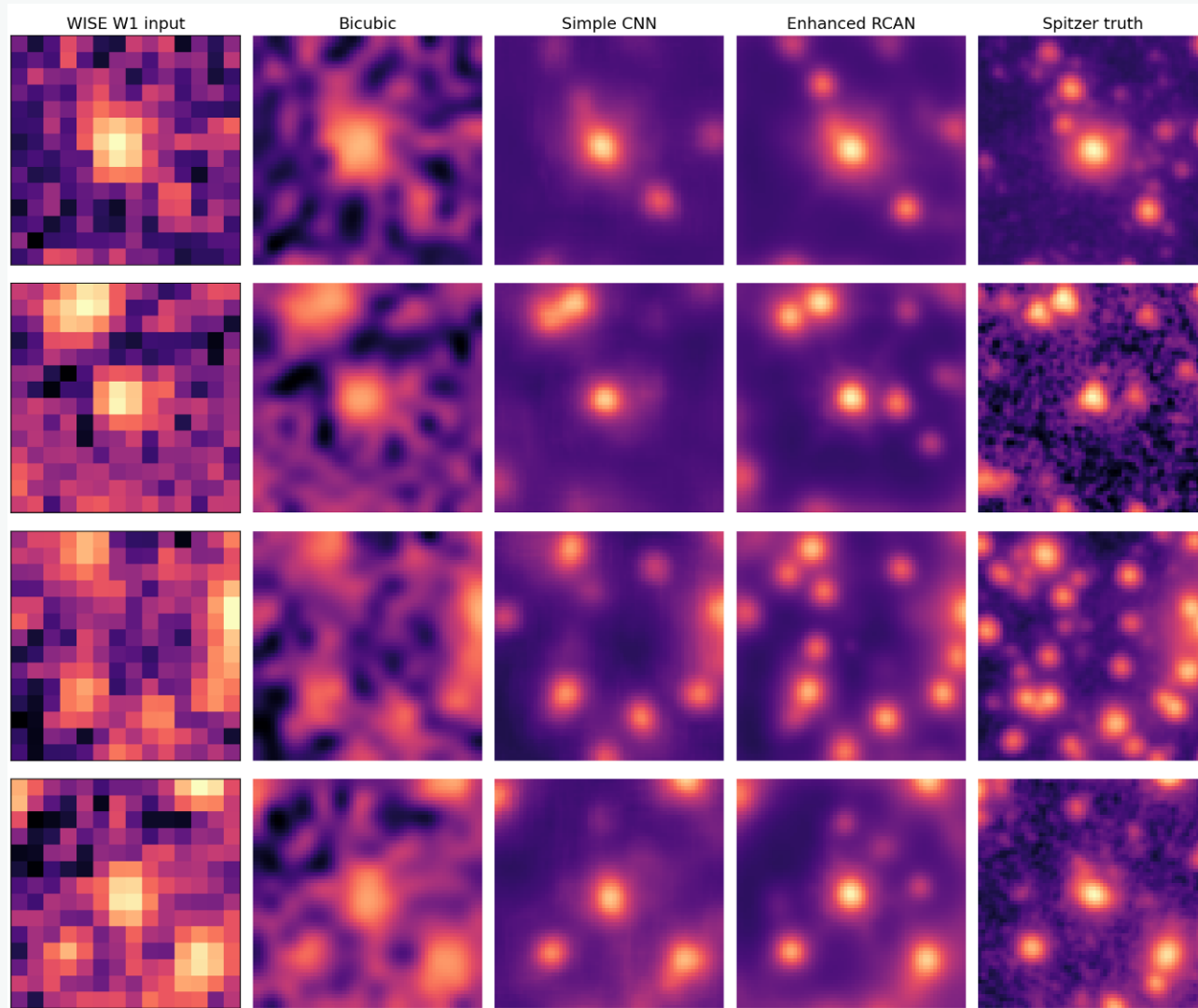
paired training cutouts

in COSMOS

The same analysis repeated for W2 → IRAC Ch2 gives consistent results; the trained model and code are public.

What 4.6× buys you

2 · WISE → SPITZER



Rezaee et al. 2026 — each row is one held-out test cutout.

ROW BY ROW

Resolved galaxy

compact source recovered where the WISE input shows a broad blob

Blended pair

the close pair is separated — the interpolation baselines render it as one broad blend

Crowded field

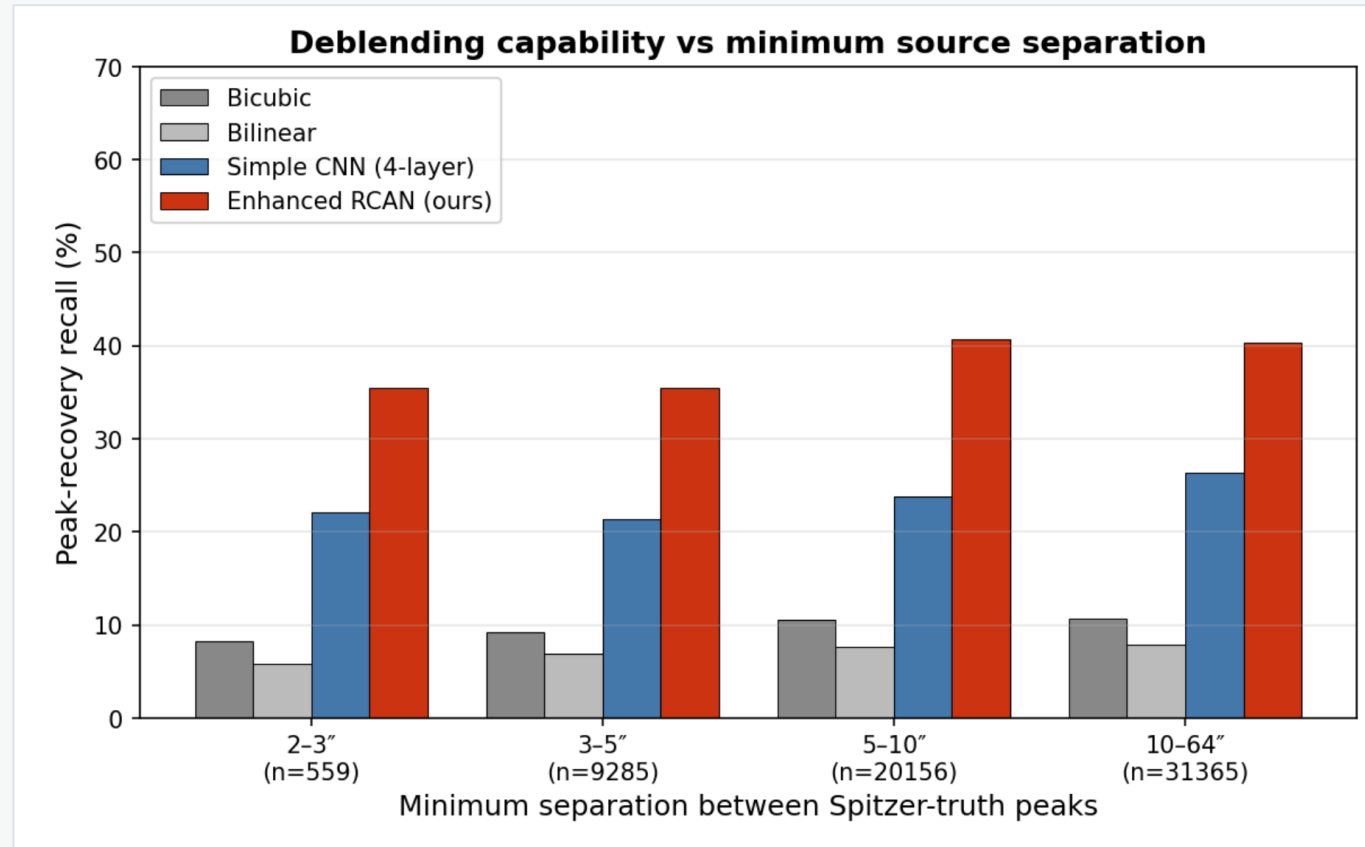
individual sources of the Spitzer truth appear; bicubic recovers only the brightest

Multi-source field

positions and relative brightnesses track the truth

Real separation, not interpolated PSF wings

2 · WISE → SPITZER



Peak-recovery recall against the Spitzer truth, binned by minimum source separation.

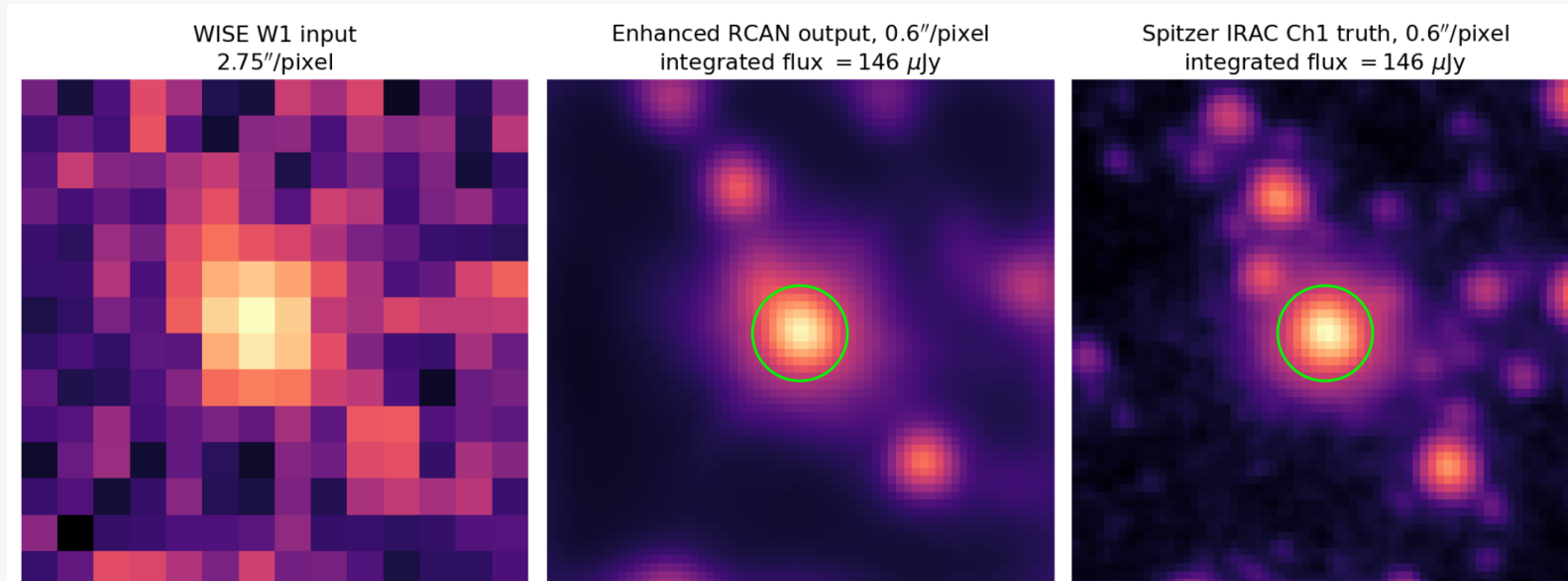
- At 3–5" — where WISE blends what Spitzer separates — the model recovers **35% of Spitzer peaks against 9% for bicubic**.
- The $\approx 4\times$ advantage holds across every separation bin, and it holds down to 2–3", below the WISE beam.
- That is the test that distinguishes genuine deblending from a smarter interpolation: interpolation cannot create a second peak where the input has one.

AND THE PHOTOMETRY FOLLOWS

Integrated aperture flux recovered to 11% median error — a factor ≈ 2 better than interpolation, and $\approx 3\times$ better on the faintest quartile.

One galaxy, end to end

2 · WISE → SPITZER



146 μJy → 146 μJy

Integrated aperture flux recovered to better than 1% for this galaxy — and the neighbouring sources in the output correspond to real sources in the truth, not to texture the model invented.

COSMOS2020 ID 376723, $z = 0.34$. Left: WISE W1 input, 2.75"/px. Middle: Enhanced RCAN output at 0.6"/px. Right: Spitzer IRAC Ch1 truth. Green circle marks the $r = 3.6''$ aperture.

11%

median integrated-flux error
across the held-out set

8% → 13%

brightest to faintest quartile —
monotonic, and predictable

all-sky

the archive this can now be
applied to, beyond the overlap

NISP sees galaxies that VIS cannot measure

3 • EUCLID → JWST

- VIS delivers weak-lensing shapes at 0.1" sampling. NISP images the same sky in Y, J, H at 0.3" — coarse enough that its shapes are not used at all.
- So a whole population — red and high-redshift galaxies, well detected in the near-infrared and absent from VIS — sits outside the lensing sample.
- Super-resolving NISP would raise the **effective source density** and extend the **source redshift distribution** — both of which strengthen the cosmological constraints directly.
- Training pairs: 18,190 NISP Y cutouts of COSMOS matched to JWST/NIRCam F115W at five times finer sampling; 4,547 held-out test galaxies.

TWO PHILOSOPHIES, TESTED HEAD TO HEAD

Conditional diffusion

Learns a generative prior. Asks: can structure that is absent from the input be plausibly synthesized?

Residual network (RRDB)

Learns a direct mapping, ESRGAN-style. Asks: how much structure is genuinely recoverable from the data that is there?

Not two architectures so much as two answers to where super-resolved information should come from.

5×

finer sampling in the
NIRCam reference

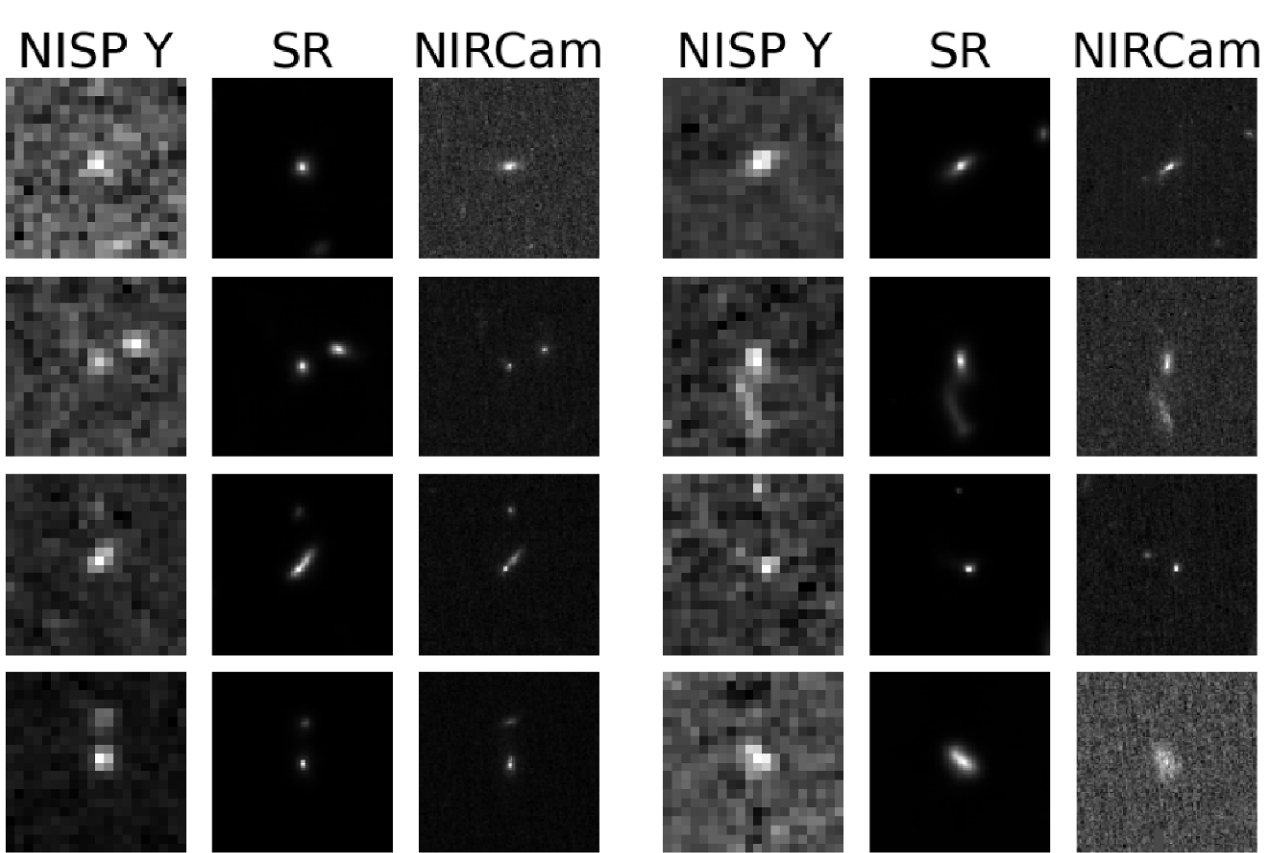
18,190

NISP–NIRCam training
pairs in COSMOS

4,547

held-out galaxies where
every number is measured

A direct mapping preserves the detail



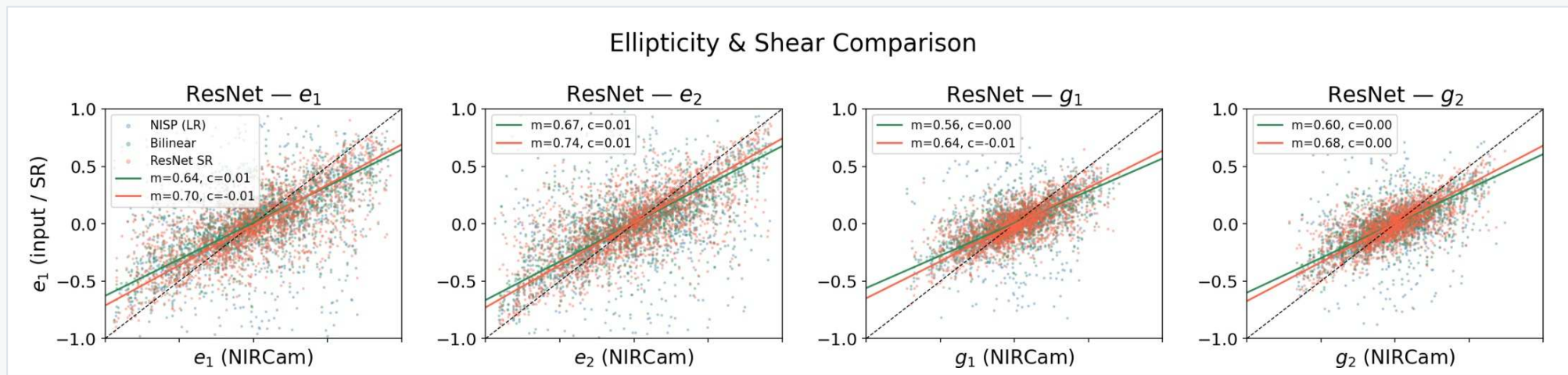
Everetts et al. 2026 — Euclid NISP Y input, ResNet super-resolution, JWST NIRCcam F115W reference.

PIXEL-LEVEL FIDELITY

The residual network outperforms both the diffusion model and a bilinear baseline on every structural metric measured against NIRCcam.

metric	ResNet	bilinear
L1 (MAE)	0.018	vs 0.032
L2 (RMSE)	0.110	vs 0.146
PSNR (dB)	37.3	vs 33.7
SSIM	0.769	vs 0.576

Does the shape information survive?



Ellipticity (e_1 , e_2) and shear (g_1 , g_2) measured with adaptive moments from the super-resolved images, against the same quantities measured from JWST NIRCams. Solid lines are orthogonal-distance-regression fits.

- The residual network recovers all four shape components in correlation with the NIRCams reference, with ODR slopes of **0.64–0.74** — closest to identity of any method tested, and well above the bilinear baseline.
- Detail-preserving regression is the promising direction for shapes; the generative prior, on this problem, is not.

ALREADY USEFUL TODAY

Morphology, deblending, target selection and strong-lens identification have a far higher tolerance for shape bias than cosmology does — and the fidelity is already there for them.

Simulation is the other half of the problem

- Shear calibration is a simulation problem: reaching Euclid's requirement of $|m| < 10^{-3}$ — and Roman's $m \sim 3.2 \times 10^{-4}$ — means analysing on the order of 10^9 galaxies.
- Analytic light profiles (Sérsic bulge + disk + knots) are interpretable and controllable, but they cannot reproduce asymmetric star-forming clumps, tidal debris, or irregular structure — exactly the morphology that biases shear.
- **So: train a diffusion model on real high-resolution galaxies, transformed into the target survey's observing regime.**

ONE RECIPE, TWO MISSIONS

The same architecture, retargeted by changing what it is trained on. Scalable in a way that GalSim-based pipelines built on analytic profiles are not.

Euclid *Scognamiglio et al. 2025*

HST/CANDELS imaging extended across Euclid's broad VIS band, then a DDPM trained on it — 32×32 px stamps, $3.2''$ on a side.

Roman HLS *Scognamiglio et al. 2026*

JWST/NIRCam GOODS-S and GOODS-N moved into the Roman regime — PSF matching, pixel-scale conversion, interloper masking that preserves correlated noise — then multi-band Y/J/H stamps.

Generated galaxies, and real ones

4 • SIMULATION

DDPM-GENERATED ROMAN-LIKE GALAXIES

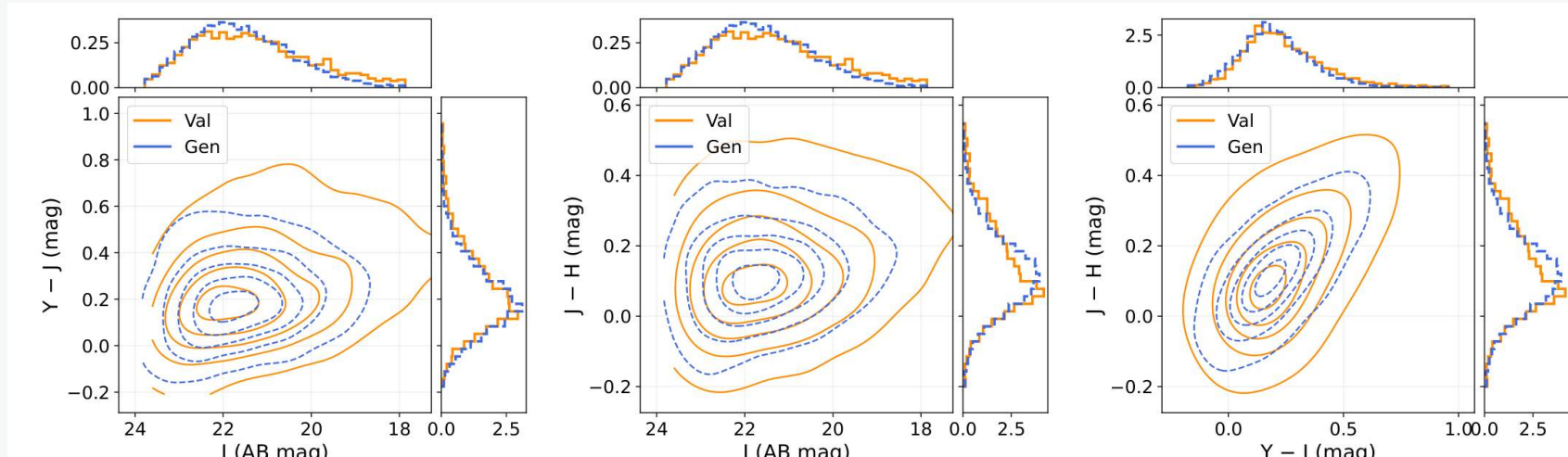


REAL ROMAN-LIKE GALAXIES (VALIDATION)



Scognamiglio et al. 2026 — RGB composites ($R = H$, $G = J$, $B = Y$), 56×56 px, $6.16''$ on a side at the Roman pixel scale. Clumps, mergers, edge-on disks and close pairs all appear on the generated side.

Not just realistic pictures – the right population



Colour-magnitude and colour-colour distributions for the validation (Val) and generated (Gen) samples, with one-dimensional marginals along each axis.

- The generated sample reproduces the marginal distributions of magnitude, size, ellipticity and peak surface brightness — and, more demanding, **their covariance structure**, in all three Roman bands.
- Two-sample KS statistics of 0.03–0.14, consistent with the bootstrap ranges expected from finite sample size.
- Agreement is strongest in the high-density core of the distributions; residual differences sit in the sparsely occupied outer contours.

WHY THE COVARIANCE MATTERS

Shear bias depends on size, brightness and ellipticity jointly. A simulation that gets each distribution right but their correlations wrong will calibrate to the wrong answer.

PART TWO

Spectroscopy

Euclid and Roman will deliver slitless spectra for millions of galaxies at low resolving power. The diagnostics we want live in lines those spectra cannot separate.

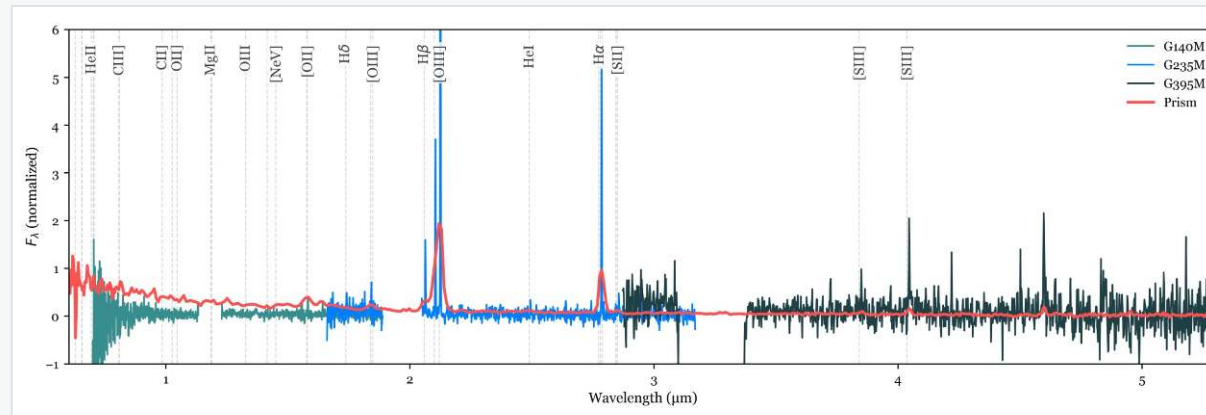


What low spectral resolution costs you

- At $R \sim 100$, $H\alpha$ merges with $[N\text{ ii}]$; $H\beta$ merges with $[O\text{ iii}] \lambda\lambda 4959, 5007$. The Balmer decrement, gas-phase metallicity, ionization state and BPT classification go with them.
- JWST/NIRSpec supplies an almost ideal training pair: the **same galaxies** observed with the prism ($R \sim 30\text{--}300$) and with the medium gratings G140M / G235M / G395M ($R \sim 1000$), over $1.0\text{--}5.3\text{ }\mu\text{m}$, from JADES.
- 2,858 paired spectra, combined into a single high-resolution reference across the full band, evaluated on a 20% held-out sample — targeting a **$\sim 10\times$ gain in resolving power, $R \sim 100 \rightarrow R \sim 1000$** .

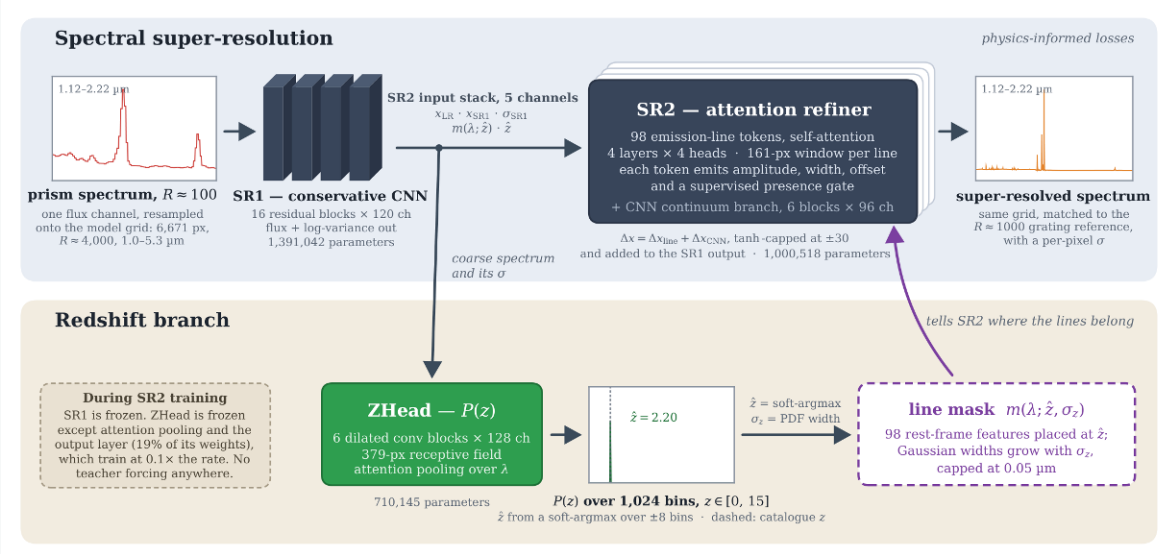
WHY IT SCALES

This is a proof of concept on JWST data — but the mapping it learns is the one Euclid and Roman grism spectroscopy needs, over millions of spectra that will never get grating follow-up.



Haghjoo et al. 2026 — one JADES galaxy at $z = 3.235$. Red: the prism spectrum. Green, blue, black: the three medium-resolution gratings. Dashed lines mark key emission lines at the galaxy's redshift.

SR2: putting the physics inside the network



The three-stage pipeline. Insets are real output for a held-out galaxy at $z = 2.198$.

SR1

a conservative CNN reconstruction with a per-pixel uncertainty (1.39 M parameters)

ZHead

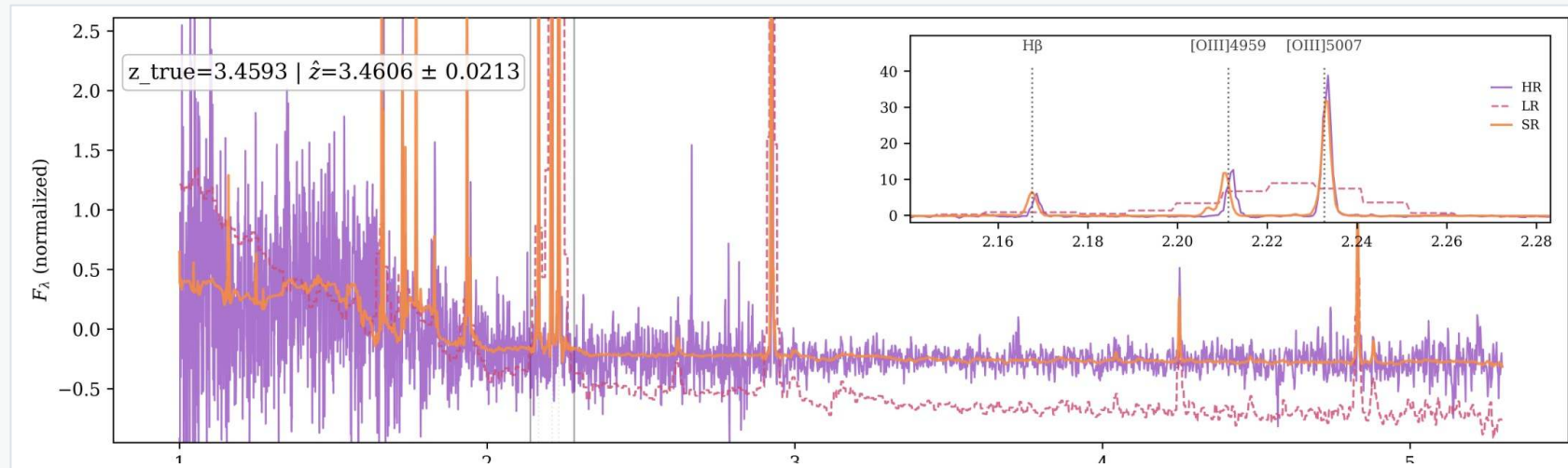
reads that coarse spectrum and returns $P(z)$ over 1,024 bins — a distribution, not a value (0.71 M)

SR2

an attention refiner: 98 emission-line tokens attending to one another, each emitting amplitude, width and position, plus a continuum branch (1.00 M)

The redshift tells the network **where the lines belong** — a soft mask that widens when the redshift is uncertain.

Lines that the prism cannot separate



A held-out galaxy at $z = 3.459$, never seen during training. Purple: the medium-resolution reference. Red dashed: the prism input. Orange: the super-resolved output. Inset zooms on $H\beta$ and the $[O\text{ iii}] \lambda\lambda 4959, 5007$ doublet.

- $[O\text{ iii}] \lambda 4959$, $[O\text{ iii}] \lambda 5007$ and $H\beta$ are fully blended in the prism input. The super-resolved output places them as **distinct peaks at the correct observed wavelengths**, tracking the grating reference.
- Residuals are noise-limited over most of the 1.0–5.3 μm range — the SR2-minus-reference residual map is close to featureless.

$\sim 10\times$

gain in resolving power

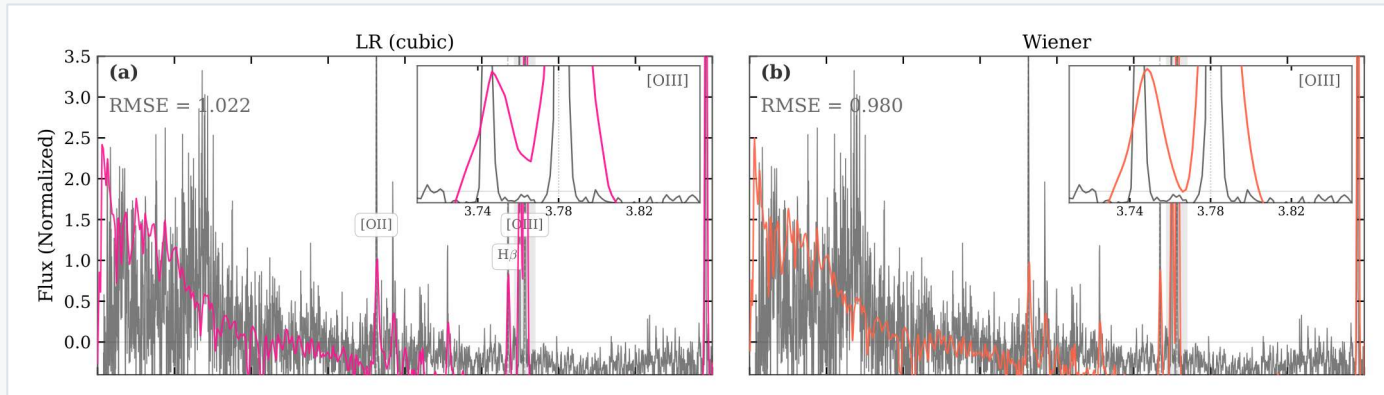
$R \sim 100 \rightarrow R \sim 1000$

The redshift returned by the middle stage is accurate enough to place the line model — which is the only thing it has to be good at.

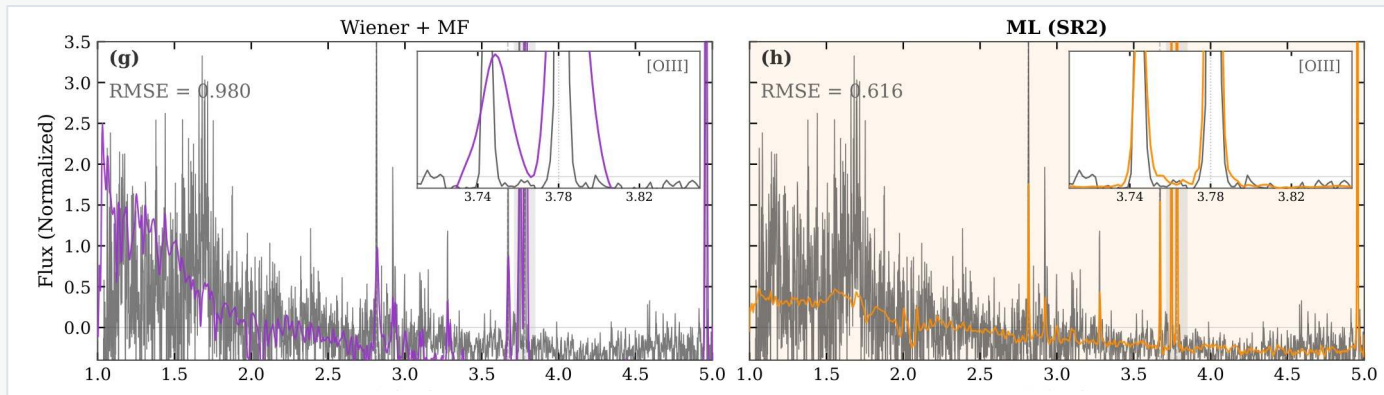
Against seven classical deconvolution methods

5 • SPECTRAL SR

CUBIC INTERPOLATION AND WIENER DECONVOLUTION



REDSHIFT-INFORMED MATCHED FILTER, AND SR2



Haghjoo et al. 2026b — one held-out JADES spectrum at $z = 6.55$. Coloured: each method. Dark grey: the high-resolution reference. Insets zoom the [O III] doublet.

- SR2 reaches **30% lower mean absolute error** than the best classical method across 1,187 held-out spectra, and the smallest line-width bias on all four diagnostic lines.
- On this spectrum: RMSE 0.616 against 0.980 for the best classical result — 37% lower.
- The redshift-informed matched filter — handed catalogue redshifts and a rest-frame line list — is statistically indistinguishable from plain Wiener. **A physical prior alone does not close the gap.**
- H α S/N is boosted by $\times 2.49$ relative to the high-resolution reference itself.

What matters in this game

LESSONS

The data, above all

The pairs are the model

Their quality, volume and diversity beat any architecture choice we have made.

Overlaps give it for free

Subaru HSC × HST · WISE × Spitzer · Euclid × JWST · prism × grating

Scale is not the lever

Our largest network here is ~3 M parameters. The deblending GAN was five convolutional layers.

Physics in the loop

Match the instruments first

PSF matching, pixel-scale transforms and noise-property preservation before the network sees anything.

Give the network what it knows

Redshift and emission-line priors, line-plus-continuum decomposition, interloper masking.

Design the loss for the science

Source-focused weighting; shape terms when shapes are the measurement.

Honest validation

Held-out sky, always

Test fields the model has never seen — and injection tests where the truth is known by construction.

Beat a real baseline

Bicubic and bilinear interpolation, and classical deconvolution, on the same data.

Population statistics, not stamps

Distributions and their covariances; and know the regime where the model is trustworthy.

None of this replaces the classical pipeline. In every project here, classical methods supplied the labels, the baselines, or both — **the learned component is a component.**

Choosing the right architecture

LESSONS

Convolutional SR networks *RCAN · RRDB / ResNet*

Direct low-resolution → high-resolution regression with residual blocks and channel attention. Faithful shapes and fluxes; no invention.

used in: **WISE** → **Spitzer** · **Euclid** → **JWST**

Generative adversarial nets *super-resolution GAN*

A generator learns what real galaxies look like; a critic forces outputs onto that manifold. Sharp, plausible structure.

used in: **Subaru HSC** → **HST deblending**

Denoising diffusion models *DDPM*

Generate galaxies by reversing a gradual noising process. Superb population realism — the right tool when you need a population, not one object.

used in: **Euclid and Roman simulations**

Physics-informed attention *multi-stage SR2*

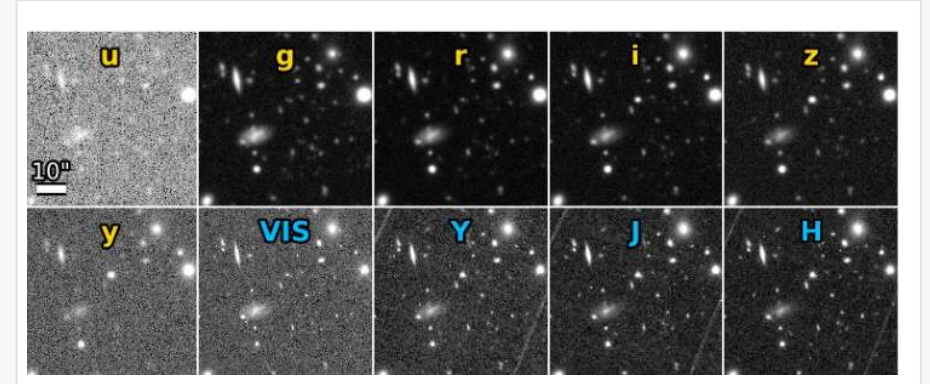
Emission-line tokens, redshift inference and parametric line profiles built into the network itself.

used in: **spectral super-resolution**

Match the model to the problem — **regression for fidelity, generative for realism, physics priors for spectra**. Not every problem calls for a foundation-scale model.

The overlaps are about to get much bigger

- Rubin, Euclid and Roman will observe overlapping sky from ~ 0.3 to $2 \mu\text{m}$ at sub-arcsecond resolution. Training pairs at **survey scale**, not deep-field scale.
- **JAISP** — a ~ 9 M-parameter foundation model that learns one shared representation of ten bands (Rubin ugrizy + Euclid VIS, Y, J, H) by masked-band reconstruction: hide one band, predict it from the other nine. No labels, no simulations, ~ 2.5 GPU-hours.
- Freeze the encoder; every measurement becomes a small head reading the same features. Train once, encode each field once, measure many times.
- The same argument, one level up: **the overlap is the prior.**



Ten bands of one matched tile: Rubin ugrizy and Euclid VIS, Y, J, H.

94% / 94%

completeness / purity vs the
Euclid MER catalogue

50 → 7-11

mas cross-survey
astrometric scatter

Ten-band fusion detects ≈ 0.3 VIS mag deeper than VIS alone, and the whole stack transfers to a second deep field with nothing retrained — **scaling is a matter of applying it, not rebuilding it.** *Hemmati et al. 2026 · github.com/xoubish/JAISP*

Takeaways

CLOSING

Survey overlaps are an empirical prior – and they are already paid for.

Every place two surveys see the same sky is a labelled training set nobody had to build.

Right-sized beats large.

A five-layer GAN, a residual channel-attention network, a 3 M-parameter spectral model. Each was chosen for what its measurement needed.

Fidelity and realism are different jobs.

Regression networks for per-object measurement; generative models for populations. Using one where the other belongs is the common failure.

Physics belongs inside the network, not only around it.

PSF matching, redshift priors, line-plus-continuum structure and science-driven losses are what separate these from generic super-resolution.

Learned priors are becoming pipeline components.

Not a replacement for decades of classical work — classical methods gave us every label and every baseline here.

THANK YOU

Questions welcome — and if you have a science case that would benefit from sharper imaging or spectra, please get in touch.

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github.com/xoubish

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